Final product audit

Meeting Actions (VIT Zero Shot Hack 2026, PS 5.3), 23 October 2026. Every result below was measured on the demo laptop (Intel Core i9-9980HK, 16 threads, macOS) unless marked otherwise. Anything not verified is labelled NOT VERIFIED, BLOCKED or NOT MEASURED.

1. Executive summary

- **Live recording works in the browser and is transcribed on this laptop.** Record audio in New meeting uses the browser's microphone APIs, shows the measured input level, plays the take back, and uploads only on *Transcribe recording*. Local Whisper returns the words in about 5 s for a 15 s clip, and the transcript is shown at once, before any language model runs. The organiser can correct lines, then start the analysis.
- **The real hardware microphone is the one thing the developers could not test**: macOS refused microphone access to the developer tools. The browser pipeline was verified with Chromium's capture device playing recorded speech. A one-minute check by the presenter (docs/LIVE_AUDIO_DEMO.md, preflight) closes this.
- **Speaker diarization is BLOCKED**: it needs Sarvam (no key on this laptop). Local Whisper labels every line SPEAKER_00, and the UI and the report say so.
- **A professional A4 PDF report** is generated from the stored records for any meeting with a transcript (0.35–0.7 s), with approval state, action register, changes to earlier commitments, outstanding questions, follow-up and transcript, in Hindi and Kannada too.
- **The UI was redesigned** (limestone, forest graphite, vermilion) with the new recording, processing and report screens; all approval and reconciliation rules are unchanged.
- **The official evaluation on three real meetings is NOT MEASURED** (no recordings received).

2. Code state

- Repository: https://github.com/sgtsujith141-wq/Zeroshot-meeting_agent
- Branch `live-audio`, fast-forwarded into `main`. Latest code commit before this document: `71bd11d`; this document is in the commit that follows it (`git log -1 -- docs/FINAL_PRODUCT_AUDIT.md`).
- Safety branch before this work: `safety/pre-live-audio-bd9bbe3` (main was `bd9bbe3`).

3. Architecture (what changed)

```
Browser (React)                          FastAPI backend                          Local services
Record audio ── MediaRecorder ──▶ POST /meetings/{id}/upload  ──▶ FFmpeg normalise ──▶ faster-whisper (small)
  (WebM/Opus, MP4/AAC, Ogg/Opus)      transcribe_only=true            │
                                       status: transcribed ◀──────────┘   transcript stored at once
Transcript ready ◀── GET /meetings/{id} (polled)
  correct a line ──▶ PATCH /segments/{key}   (new revision)
  Analyze meeting ──▶ POST /process {interpret, minutes} ──▶ Ollama qwen3:8b ──▶ rules ──▶ proposal
Results / report ──▶ GET /meetings/{id}/report.pdf ──▶ HTML from stored records ──▶ headless Chromium ──▶ A4 PD
```

- **Transcript first.** `run_pipeline(steps=("transcribe",))` stops after speech-to-text with status `transcribed`. Each step commits on its own, so a later model failure never removes the transcript; `POST /process` with interpret+minutes retries without transcribing again.
- **Audio-only WebM** (what Chromium records) is stored as audio, not video (ffprobe check).

- **Whisper pre-load.** The model loads in a background thread at server start (`WHISPER_PREWARM=0` disables it).
- **Report.** `services/report.py` builds print-ready HTML from the database and renders it with headless Chromium (`--print-to-pdf`, CSS page margin boxes for running headers and page numbers). System fonts only: nothing is bundled. Endpoints: `GET /api/meetings/{id}/report.pdf` (`?download=1` for an attachment) and `report.html`, both organiser-only.

4. Speech provider configuration

Provider	State on this laptop
Local Whisper (`faster-whisper-small`, CPU, int8)	**Active.** Installed and verified
Sarvam saaras:v4 Batch (diarization)	**BLOCKED**: `SARVAM_API_KEY` not set. Implemented, never run live
Sarvam short-audio REST API	Not implemented. It has no diarization; with no key to verify it, it was not added
Gemini (LLM)	Not configured (no key). The local Ollama `qwen3:8b` is the active model

5. Live microphone test results

Test	Result
Chromium: Record → level meter → Stop → Play → Transcribe → transcript → correction, real Whisper (`e2e/flows/live-audio.spec.ts`; capture device plays a 12 s TTS speech WAV)	**PASS**, 3 of 3 runs; transcript visible **6.5–6.6 s** after *Transcribe recording* (includes the model load; sandboxes do not pre-load)
Measured input level drives the meter (bars above the floor while speech plays)	**PASS** (same test)
Server accepts and transcribes what each browser records: Safari AAC `.m4a`, Firefox Opus `.ogg`, Chrome Opus `.webm`	**PASS**, 4.8 s each for 14.7 s, correct words
Silent recording	Shows "No speech recognised" (no invented text)
A person speaking into this laptop's built-in microphone	**BLOCKED for the developers**: macOS denied microphone access to the developer tools ("Cannot use MacBook Pro Microphone"). The presenter must run the preflight in docs/LIVE_AUDIO_DEMO.md
Safari / Firefox recording in an automated browser	NOT VERIFIED (no matching automation build); formats verified server-side only. Chrome and Edge are not installed on this laptop

6. Multi-speaker status

BLOCKED. Diarization requires the Sarvam Batch API and a key. With local Whisper every segment is `SPEAKER_00`; the New meeting page, the transcript panel and the report state that speaker separation is not available. Speakers can be named afterwards (Speakers tab); a label is never treated as a verified identity.

7. Audio and language limitations

- Timestamps are per Whisper segment, not per word.

- `faster-whisper-small` on code-mixed speech: on the synthetic TTS check, a Kannada line came out phonetically in Devanagari (docs/evaluation.md). Sarvam is the intended provider for code-mixed meetings.
- Recordings are limited to 30 minutes in the browser and to `MAX_UPLOAD_MB` (200 MB) on the server.
- The analysis with the local model takes minutes (below); only transcription is fast.

8. Screen-by-screen changes

Screen	Change
Whole app	Limestone canvas `#F5F2EB`, white surfaces, deep ink text, forest graphite `#172D30` sidebar, vermilion `#C8583D` marks. Buttons and links use `#AB452E` and secondary text `#5A676E` so text meets WCAG AA (white on `#C8583D` is 4.27:1)
Intro	Palette only; plays on every load (presenter's request); Esc / Skip intro
Overview	"Meeting Actions – From meetings to work that gets done.", *Record or upload a meeting*; Meetings awaiting review (including transcripts waiting for analysis), Upcoming tasks, Recently processed meetings
New meeting	Record audio / Upload recording / Upload transcript; the recorder dominates in record mode; time zone under More options
Processing	Five stages from real status and log times: Audio received, Transcription, Transcript ready, Extracting decisions and actions, Preparing meeting results; transcript below as soon as it exists; Retry analysis; Try transcription again
Results	Summary, discussions, decisions, action items, questions; *Review and approve tasks*, *Download meeting report*, *Preview report*
Approvals, Tasks, Ask AI, Follow-ups	Restyled by the palette; behaviour unchanged from the previous release (before → after lines, compact task list, fast task answers)

9. PDF report validation

Case	Pages	Render	Checked
Approved meeting (demo meeting 1)	3	0.35 s	APPROVED tag; tasks with board status " (approved)"
Cross-meeting changes (meeting 2)	3	0.37 s	Owner Priya → Samiksha, due date added, "Yes" approved
Code-mixed, pending (meeting 3)	3	0.39 s	DRAFT: PENDING APPROVAL; "Proposed, not approved"
Unresolved clarifications (meeting 4, audio)	3	0.41 s	4 open questions; Devanagari lines shaped; speaker-separation notice
Short English recording (a synthetic TTS clip uploaded as WebM; Whisper, then Ollama)	3	0.39 s	summary, Stripe decision, two actions, one open question
Long transcript (192 segments, synthetic, not analysed)	10	0.49 s	TRANSCRIPT ONLY; all 192 segments present (pdftotext); no empty first page

Pages were rendered to images (pdftoppm) and inspected. Defects found and fixed during inspection: running header missing the title; proposed actions coloured as approved; a transcript-only report starting with an almost empty page; transcript-named speakers listed as "not identified". `tests/test_report.py` checks the approval wording, escaping, Unicode, access control, and (where Chromium and poppler exist) the PDF text against the records.

10. Test results

Suite	Tests	Result
Backend `pytest` (incl. `test_transcribe_first.py` with real Whisper on a WebM, and `test_report.py` with a real PDF render read back by pdftotext)	123	**123 pass**
Browser suite `e2e/flows` without the 4 live-model tests, one clean run: navigation, approvals and reconciliation, questions, meetings, tasks, Ask AI, follow-ups, links, accessibility (axe, WCAG 2.1 A/AA, both themes, intro), final UI, **live recording**	101	**101 pass** (5.5 min)
Live-model suite `slow.spec.ts` (Ollama: upload → analysis → re-run → approve; grounded answer with passages shown while the model writes; model down; live test)	4	**4 pass** (5.0 min)
Screen checks `screens.spec.ts` (every route: no horizontal overflow, no control outside the viewport) at 1440, 1024, 390 px light and 1440, 390 px dark	5 runs	**5 pass**
Read-only smoke tests against the live demo server	4	**4 pass**
Frontend production build; CI	–	passes; CI: see the push

During this work one browser test failed once for a test-side reason: a second page opened with `browser.newPage()` did not skip the intro. It was fixed and the whole suite then passed in one run.

11. Performance (measured)

Measurement	Result
Recording duration in the browser test	~12 s of speech
Upload + transcription, 14.7 s clip (Safari/Firefox/Chrome formats)	4.8 s each
Transcription only, 14.7 s, model loaded (`small`)	4.6–5.0 s (`base` : 1.7 s, not adopted: weaker on accents and code-mix)
Whisper model load	4.4 s (now done at server start)
Click *Transcribe recording* → words on screen (browser test, includes model load)	6.5–6.6 s
Full analysis after transcription (4-line transcript, Ollama `qwen3:8b`)	139.2 s (interpretation 74.6 s + minutes 64.0 s)
PDF render	0.35–0.7 s; 3 pages typical, 10 pages for 192 segments
Ask AI, task-state question (from the previous release)	median 4.0 ms; model answers 18–40 s

12. Remaining issues

- Hardware microphone with a human speaker: not verified by the developers (operator check).
- Diarization: needs a Sarvam key.
- The analysis with the local model takes about 2 minutes even for a short clip.
- Safari/Firefox recording not automated; Chrome/Edge not installed on the demo laptop.
- The live demo database contains duplicate tasks (TASK-5/6) from a hand-approved "TEST" meeting; it was left untouched.
- A screen-reader walkthrough was not done.

13. Known risks

- A browser or OS microphone permission prompt during the talk: do the preflight.
- Wrong input device (an iPhone or Teams virtual microphone is also present on this laptop).
- Ollama not running: the transcript still shows; the analysis fails with a clear message and can be retried.

14. Demonstration readiness

NEEDS FIXES, one operator check. Everything the developers could run passes. The live judge demo depends on one thing they could not run: a person speaking into this laptop's microphone in the presentation browser (Safari or Firefox; Chrome is not installed). The preflight in docs/LIVE_AUDIO_DEMO.md takes about a minute (grant the microphone, record 10–20 s, see the words). Once it passes, the live route is ready, with the transcript in seconds and the analysis in about 2 minutes, and the labelled earlier sample as the fallback.

15. Outstanding official submission requirements

- The evaluation on three real meetings: **NOT MEASURED** (no consented recordings received; procedure in `evaluation/real_eval.py`).
- Gemini, Sarvam and SMTP: **not verified live** (no keys or server).
- No release tag (`v1.0.0-zeroshot`) has been created, because the items above are open.